

SIMATS ENGINEERING
C03 ASSESSMENT TOOL1

COURSE CODE: CSA1610

COURSE NAME: DWDM

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EXPERIMENTS

Q1 R Code for Covariance and Correlation:

CODE:

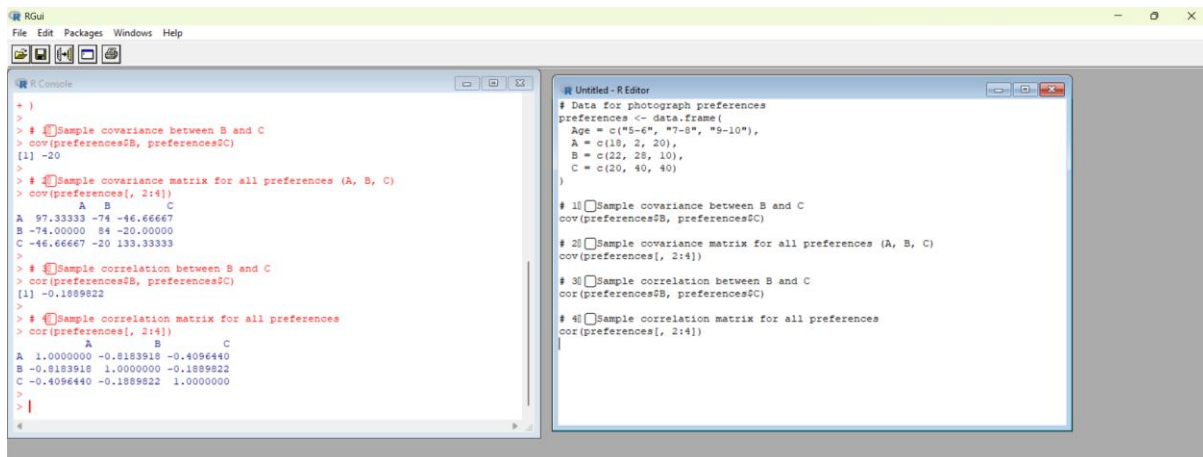
```
# Data for photograph preferences
preferences <- data.frame(
  Age = c("5-6", "7-8", "9-10"),
  A = c(18, 2, 20),
  B = c(22, 28, 10),
  C = c(20, 40, 40)
)

# Sample covariance between B and C
cov(preferences$B, preferences$C)

# Sample covariance matrix for all preferences (A, B, C)
cov(preferences[, 2:4])

# Sample correlation between B and C
cor(preferences$B, preferences$C)

# Sample correlation matrix for all preferences
cor(preferences[, 2:4])
```



Q2 R Code for Boxplot and Outlier Detection

CODE:

Data: Points scored by players

```
points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)
```

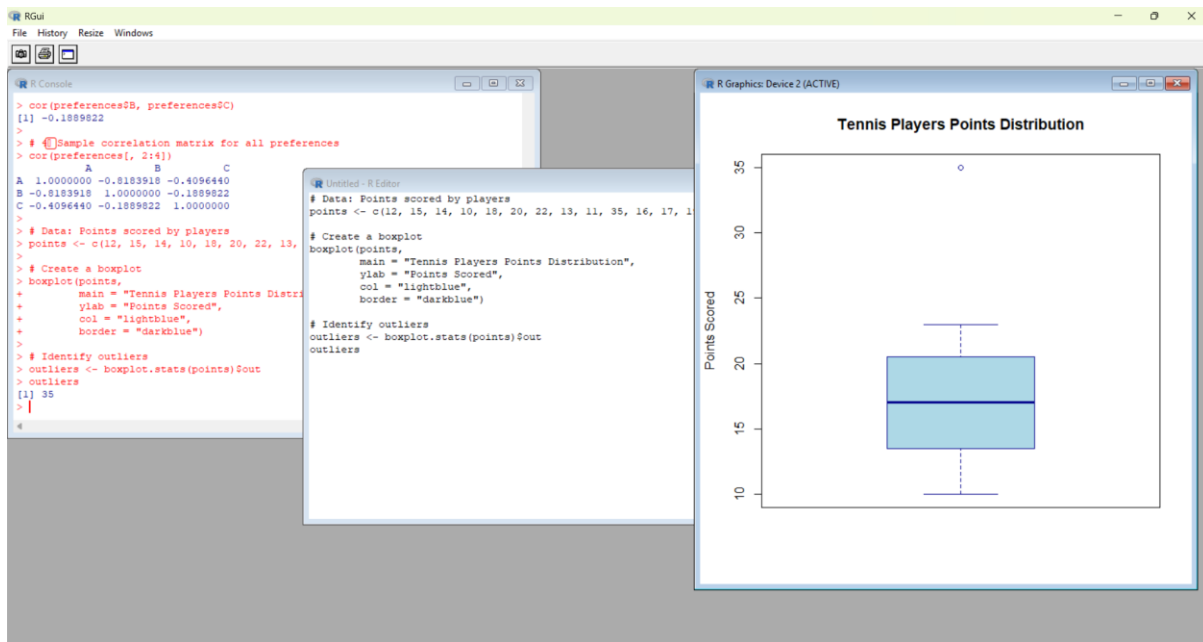
Create a boxplot

```
boxplot(points,
  main = "Tennis Players Points Distribution",
  ylab = "Points Scored",
  col = "lightblue",
  border = "darkblue")
```

Identify outliers

```
outliers <- boxplot.stats(points)$out
```

outliers



Q3 R Code for Decision Tree Classifier

CODE:

```
# Install and load required packages
```

```
# install.packages("rpart")
```

```
# install.packages("rpart.plot")
```

```
library(rpart)
```

```
library(rpart.plot)
```

```
# Dataset
```

```
loan_data <- data.frame(
```

```
  Age = c("young","young","middle","old","old","old","middle","young","young","old",
          "middle","middle","old","young","middle"),
```

```
  Income = c("high","high","high","medium","low","low","low","medium","low","medium",
             "medium","high","medium","medium","low"),
```

```
  Employment = c("employed","self-
employed","employed","employed","unemployed","self-employed",
```

```
                "unemployed","employed","unemployed","self-employed","employed","self-
employed"),
```

```

      "unemployed","self-employed","employed"),
Credit = c("good","average","good","good","poor","average","poor","average","poor","good",
      "average","good","poor","average","poor"),
Loan = c("yes","yes","yes","yes","no","no","no","yes","no","yes","yes","yes","no","yes","no")
)

```

Build Decision Tree

```

tree <- rpart(Loan ~ Age + Income + Employment + Credit, data = loan_data, method =
"class")

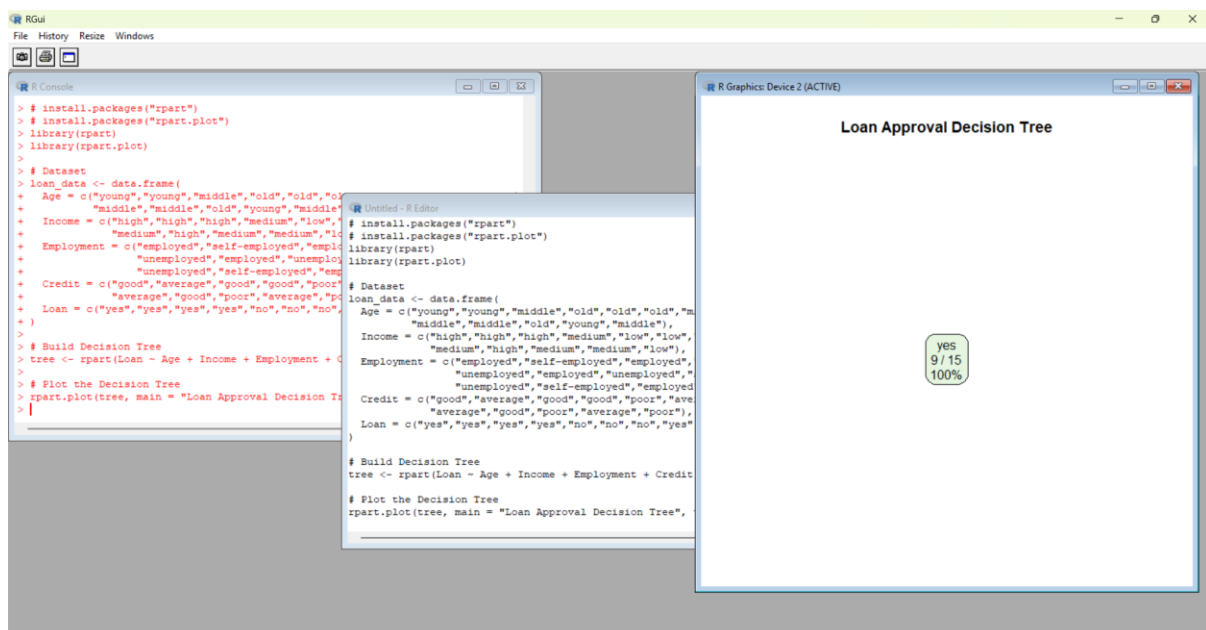
```

Plot the Decision Tree

```

rpart.plot(tree, main = "Loan Approval Decision Tree", type = 3, extra = 102)

```



Q4 R Code for Mean, Median, Range, and Boxplot

CODE:

Data for Class A and Class B

```

classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)

```

```

classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)

```

```
# Mean, Median, Range
```

```
meanA <- mean(classA)
```

```
medianA <- median(classA)
```

```
rangeA <- range(classA)
```

```
meanB <- mean(classB)
```

```
medianB <- median(classB)
```

```
rangeB <- range(classB)
```

```
# Display results
```

```
meanA; medianA; rangeA
```

```
meanB; medianB; rangeB
```

```
# Boxplot comparison
```

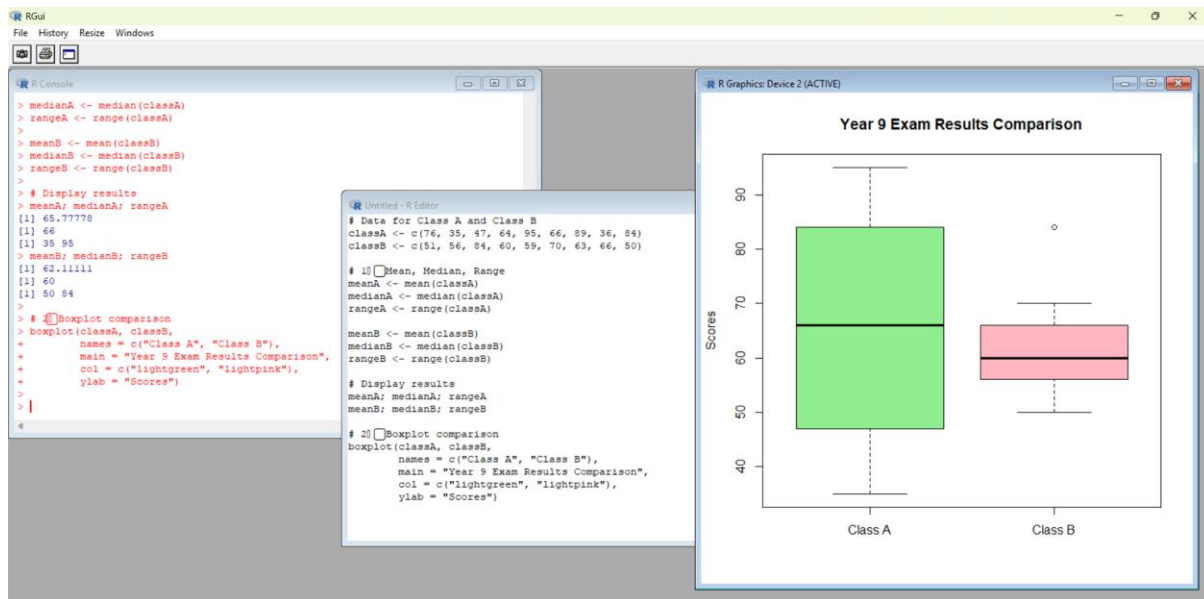
```
boxplot(classA, classB,
```

```
  names = c("Class A", "Class B"),
```

```
  main = "Year 9 Exam Results Comparison",
```

```
  col = c("lightgreen", "lightpink"),
```

```
  ylab = "Scores")
```



Q5 R Code for Statistical Analysis

CODE:

Data

```
age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61)
```

```
fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2, 34.6, 42.5, 28.8, 33.4, 30.2,
34.1, 32.9, 41.2, 35.7)
```

(a) Mean, Median, Standard Deviation

```
mean(age); median(age); sd(age)
```

```
mean(fat); median(fat); sd(fat)
```

(b) Boxplots

```
par(mfrow = c(1, 2)) # Display side-by-side
```

```
boxplot(age, main = "Boxplot of Age", col = "lightblue", ylab = "Age")
```

```
boxplot(fat, main = "Boxplot of Body Fat %", col = "lightgreen", ylab = "% Fat")
```

(c) Scatter Plot and Q-Q Plot

```
par(mfrow = c(1, 2))
```

```
plot(age, fat, main = "Scatter Plot: Age vs Body Fat", xlab = "Age", ylab = "% Fat", pch = 19, col = "darkblue")
```

```
qqplot(age, fat, main = "Q-Q Plot: Age vs Body Fat", xlab = "Age Quantiles", ylab = "Body Fat Quantiles", col = "purple")
```

